
Index

- 21st century, xi, 1, 15, 240
- acceleration, 139
- accuracy calculations, 27, 126, 141
- accuracy formula, 114, 139, 148
- α , 146–148, 250
- apparent error, 216
- atom, 217, 228, 229
- batting averages, 7, 8
- BAX gene, 163, 164
- Bayes, x, 4, 5, 10, 12, 14, 18, 19, 23–25, 28, 55, 57, 69, 82, 90, 95, 97, 192, 205, 217, 220, 223, 231
- hierarchical, 13
- objective, 13
- Bayes factor, 45, 74
- Bayes intervals, 233
- Bayes methods, 2
- Bayes rule, 2, 3, 7, 13, 18, 53, 189, 211, 212, 230, 232, 233
- Bayes theorem, 2, 27, 80, 188, 199, 217
- Benjamini and Hochberg, x, 18, 23, 28, 46, 48, 52, 69
- Benjamini–Hochberg algorithm, 48, 49, 54, 56, 57, 59, 64, 66, 69, 173, 204
- bin centers, 116, 127, 129
- bin width, 79, 90, 116, 118, 119
- binomial distribution, 24, 31, 48
- bivariate normal density, 117
- block statistic, 150
- block test, 152
- Bonferroni bound, 17, 35, 36, 38, 39, 43, 81, 202
- Bonferroni's method, 81, 200
- Boole's inequality, 35, 40
- bootstrap, 113, 115, 138–140
- non-parametric, 115, 138, 139
- bootstrap methods, 112, 138
- brain tumor study, 207
- BRCA data, 152, 162, 167, 168
- canonical parameter, 217, 243
- cardio data, 149–151, 153, 162, 249
- central counts, 98, 160, 161
- central matching, 97, 99–101, 104, 105, 108, 112, 129–131, 226, 227
- estimates, 98–100, 103, 104, 129, 130, 226
- chi-square data, 92, 93, 107, 112, 206, 249
- chi-square distribution, 4, 93
- chromosome, 147, 148, 202–205
- sites, 202, 203
- Clemente, 7–9
- coefficient of variation, 23, 24, 77, 128, 178, 236
- column permutation, 153, 165, 166, 168
- combination, 28, 120, 163, 184–186, 192, 209
- comparability, 94, 185, 206–209
- conditional probability, 2, 188, 197, 228
- confidence interval, 12, 14, 24, 230–233
- control algorithm, 48, 173, 204, 230, 231, 233
- Corbet, Alexander, 233–237, 241
- Cornish–Fisher, 59, 135, 139
- correlation, x, 24, 26, 27, 51, 55, 56, 68, 77, 104, 106, 111, 114, 115, 117, 118, 128, 129, 141, 143, 146–150, 152, 153, 155, 157, 159–162, 165, 167, 174, 176–178, 183, 220, 222, 223
- column, 139, 143–146, 152, 153, 156, 168
- count, 159
- ecological, 149
- negative, 68, 111, 160, 162

- rms, 55, 56, 104, 106, 113, 122, 123, 141, 145–149, 153–157, 159, 176, 250
- row, 111, 143–147, 153, 156, 168
- row and column, 141–143
- sample, 67, 142–144, 222
- simultaneous tests of, 64, 67
- correlation across cases, 106, 108, 111, 113
- correlation across units, 106, 109, 110
- correlation clusters, 68
- correlation corrections, 221
- correlation distribution, 118, 120
- correlation effects, 109, 139, 151, 159, 162, 176
- correlation matrix, 113, 117, 144, 149, 152, 222
- correlation of *t*-values, 157
- correlation structure, 42, 56, 174, 183, 221
- count vector, 116, 117, 126, 159, 161, 196
- covariance matrix, 113, 116–119, 125, 128, 142, 148, 149, 154, 155, 157, 167, 221, 236, 246
- covariate, 11, 80, 95, 107, 133, 147, 197, 198, 202, 206
 - unobserved, 107, 108, 111
- CRAN, 174, 215
- cross-validation, 215, 216, 219, 221–223, 240, 250
- cumulant, 218, 230, 244
 - generating function, 243, 244
- cumulative distribution functions (cdf), 16, 18, 19, 32, 39, 41, 43, 51, 53, 54, 60, 68, 72, 91, 93, 105, 111, 113–115, 119, 125, 132–135, 138, 165, 169, 173, 237
 - empirical, 53, 55, 124, 234
 - non-null, 86
- deconvolution, 218
- delta-method, 23, 115, 126, 127, 130
- demeaning, 154, 155
- deviance function, 244, 246
- dichotomous case, 212
- diffusion tensor imaging, 26
- Dirichlet, 201, 245
- disease status, 202
- DTI data, 26, 28, 32, 34, 54, 57, 59–61, 71–73, 181, 182, 185, 186, 189, 199, 249
- dyslexic, 26, 34, 181, 183
 - and normal children, 26, 193
- Ebay, 215, 218–223, 240, 250
- EER, 43, 45
- effect size, 181, 212–214, 217–220, 223–225, 227–231, 233, 250
 - estimation, 211, 212, 227, 230, 241
- effective sample size, 153, 155–157
- efficiency, 1, 27, 90, 195
- eigenvalues, 142–144
- eigenvector, 149, 151, 152
- empirical Bayes, ix, x, 1, 4, 5, 7, 10, 12–14, 18, 20–22, 27–29, 37, 46, 51–53, 55, 57, 63, 66, 69, 82, 88–90, 114, 115, 190, 191, 198, 199, 212, 213, 217–220, 223, 229, 231, 233–235, 240, 241, 250
 - confidence intervals, 12, 14, 233
 - methods, 7, 17, 30, 37, 70, 113, 206, 209, 240
 - posterior band, 229
- empirical distribution, 20, 79, 109, 139, 169
- empirical null, 84, 90, 91, 94–98, 106–108, 111, 112, 130, 141, 163, 167, 169, 183, 184, 186, 194, 224–226, 250
 - density, 90, 164
 - distribution, 56, 89, 160, 226, 227
 - estimation, 97, 102, 194, 196
- empirical probability distribution, 29
- enrichment, 82, 163, 164, 167, 170, 171, 173, 174, 178, 181–185, 191, 192, 194
 - analysis, 163–165, 184
- estimation, x, xi, 1–3, 10, 13, 14, 20, 27, 28, 57, 58, 60, 71, 75–77, 80, 82, 88–90, 98, 99, 103, 104, 123, 156, 188, 195, 201, 220, 229, 233, 241, 243
 - effect size, 211, 212, 227, 230, 241
 - empirical Bayes, 1, 4, 13, 63, 113, 240
 - empirical null, 97, 102, 194, 196
 - James–Stein, 10, 11, 209
 - maximum likelihood, x, 1, 74, 75, 238, 245, 246
- estimation accuracy, 113, 148
- exchangeability, 80
- expected false and true positives, 82, 83
- exponential family, xi, 74, 82, 103, 104, 171, 174, 217, 224, 230, 243–247

- multiparameter, 105, 137, 245, 247
 - one-parameter, 137, 241, 243
- exponential tilting, 243, 244
- expression matrix, 47, 61, 133, 144, 146, 149
- false coverage rate, 230–232
- false discoveries, 21, 22, 47, 51, 59, 232
- false discovery proportion, 22, 23, 29, 47–49, 51, 53, 55, 56, 106, 107, 130, 160, 201
- false discovery rate, *x*, *xi*, 18, 20, 22, 23, 26, 28, 30, 37, 42, 43, 45–47, 50, 53, 57, 58, 60, 62, 64, 69, 70, 72, 73, 81–83, 97, 102, 105, 106, 128, 159, 160, 164, 184–186, 188, 194, 197–201, 205, 208, 212, 223, 225
 - Bayes, 18, 19, 23–25, 29, 55, 200
 - local, 19, 28, 29, 57, 70, 73, 77, 78, 82, 83, 86–88, 90, 97, 183, 185, 187, 197, 223, 226, 229, 250
 - positive, 51
- false negative rate, 57, 69
- FCR control, 230, 231, 233
- FDR control, 30, 37, 45, 46, 48, 50, 51, 57, 58, 69, 70, 173, 204, 231, 233
- filtration, 109
- Fisher, R.A., ix, *x*, 14, 15, 31, 32, 36, 45, 48, 65, 111, 112, 234, 237–239, 241
 - Fisher's linear discriminant function, 211, 212, 214
 - Fisher's prediction formula, 237
 - Fisher's scale of evidence, 31, 46
- fold-change, 207–209
- frequentist, ix, *x*, 1, 2, 5, 10, 14, 15, 18, 30, 37, 48, 54, 115, 199, 200, 205, 230, 233
 - methods, 205, 233
- FWER, 30, 34, 35, 43, 44, 47, 49, 50, 81, 83, 185, 199, 200, 205
 - control, 35, 37, 40, 43
- gamma density, 237
- gamma distribution, 134
- gene expression, 15, 141, 149, 163, 167, 212
- gene set, 163–171, 173, 174, 176, 181, 183, 184
- generalized linear model (GLM), 75, 127, 247
 - Poisson, 131, 247, 248
- GSEA statistic, 165, 172–174
- Hermite polynomial, 118, 119
- hierarchical model, 241
 - Bayes, 27, 28, 96, 182
 - normal, 217, 223–226, 228, 241
 - Poisson, 240
- high-dimensional statistical inference, 1
- higher criticism, 59, 69, 205
- hippocampus, 197
- HIV data, 94, 96, 99, 105, 110–112, 148, 249
- Holm's procedure, 36–38, 40, 81
- homogeneity assumption, 123–125
- hyperparameter, 13
- independence, 24–26, 36, 39, 50, 55, 56, 59, 66, 76, 84, 92, 106, 111, 113, 124, 125, 128, 130, 141, 152, 153, 221, 222, 235, 236
 - assumption, 24, 26, 55, 60, 106
 - column-wise, 141, 149, 153, 162
- indirect evidence, 7, 229, 240
- influence function, 127, 129, 130, 139
- James–Stein estimator, 1, 5–7, 14, 27
- Jensen's inequality, 22, 23, 155
- k*-FWER, 43–45
- kidney data, 11, 14, 249
- Kolmogorov–Smirnov test statistic, 165, 172
- Kronecker product, 154
- Kullback–Leibler distance, 244
- kurtosis, 132, 135
- large-scale inference, 3, 9, 34, 163
- learning from the experience of others, 10, 27, 184, 206, 229, 240
- least squares, 10, 12, 110, 129, 131, 245
- Lehmann alternatives, 20
- leukemia data, 63, 64, 98, 111–114, 124, 132, 138, 146, 147, 224, 225, 249
- likelihood, *x*, 1–3, 5, 13, 74, 75, 102–104, 171, 238, 245, 246
- limma, 184
- Lindsey's method, 74–76, 88, 247
- linear logistic model, 194, 195
- linear regression, 5, 11, 34, 150, 152, 240
- local averaging, 181–183
- local true discovery rate, 84
- locfdr, 34, 75, 76, 82, 86, 101, 103, 104, 127, 153, 226, 250
- logistic regression estimate, 189
- Malaysian butterfly data, 233, 234, 236
- marginal density, 18, 19, 123, 187, 188, 199, 217, 218, 224, 233

- marginal distribution, 2, 4, 177
- max- T , 40–43
- maximum likelihood estimator (MLE), 3, 5, 102
- maxmean statistic, 171–173, 184
- mean, 2, 8, 9, 24, 28, 56, 68, 77, 87, 98, 104, 107, 115–117, 121, 125, 126, 132–135, 137, 141–144, 147, 149, 154, 156–159, 161, 165–168, 172, 173, 177, 178, 184, 211, 214, 250
 - and variance, 6, 23, 60, 82, 114, 144, 158, 159, 161, 180, 228, 230, 244
- posterior, 13, 218, 223
- Mehler's identity, 118, 121
- Michigan lung cancer study, 222, 223, 241
- microarray, ix, 1, 15, 27, 28, 43, 45, 61, 81, 91, 96, 106, 107, 109, 121, 141, 143, 145, 148–150, 152, 163, 164, 167, 181, 206–208, 212, 215
- microarray experiment, 61, 107, 112, 133, 143, 152, 163, 223
- microarray study, 27, 61, 64, 91, 109, 113, 148, 163
- Miller, Rupert, xi, 30, 45, 186, 205
- min- p , 40, 41
- missing mass problem, 239
- missing species problem, 233, 237, 241
- mixture density, 18, 21, 29, 71, 74, 88, 96, 101, 114, 187, 250
- MLE method, 1, 90, 99, 101, 102, 104, 105, 108, 112, 153, 183, 189
- multi-class model, 117, 119, 187
- multinomial, 118, 125, 161, 236, 245, 247
- multivariate normal, 2, 6, 28, 36, 146, 211, 245
 - calculations, 153, 180
- natural parameter, 137, 217, 243, 244, 247
- natural spline basis, 76, 90
- nearest shrunken centroid, 215, 240
- negative binomial, 237, 238
- new species, 234, 235, 237, 238
- non-central χ^2 , 178, 180
- non-central t , 132, 137, 221
 - distribution, 133
- non-null counts, 84, 85
 - smoothed, 84, 85, 96, 124
- normal distribution, 3, 16, 84, 139, 153, 213
 - matrix normal, 154
 - multivariate normal, 211
- null and alternative densities, 29
- null hypothesis, x, 16, 26, 31, 32, 48, 49, 51, 55, 58, 65, 74, 83, 89, 96, 97, 105, 111, 113, 132, 135–137, 141, 149–151, 164, 165, 170–172, 174, 194, 195, 201, 224, 227, 250
 - blurry, 97
 - complete, 40–42
 - enrichment, 194
- null proportion, 20, 28, 49, 54, 60, 66, 90, 98
- odds ratio, 2, 211, 227
- optimal discovery procedure, 45
- p -values, 31–41, 45, 48–53, 57, 66, 70, 136, 141, 167, 169, 173, 174, 200, 204
 - adjusted, 34–37, 39–42, 202
- p53 data, 163, 166, 173, 184, 194, 250
- P53.UP, 194, 195
- pamr, 215–219, 240
- PCER, 43
- penalty, 5, 50, 76, 201, 204, 205
 - correlation, 114, 118, 119, 122, 123, 128
- permutation algorithms, 39
- permutation calculations, 40, 97, 109, 110, 177–180, 183
- permutation distribution, 41, 62, 64, 89, 111, 169, 180
- permutation methods, 41, 42, 110–112, 149, 171, 174
- permutation null, 97, 109, 111, 141, 149
 - distribution, 64, 109, 110, 184
- point estimate, 22
- Poisson assumption, 25, 170
 - Poisson-independence, 24, 25, 58, 200
- Poisson deviance, 76, 248
- Poisson process, 234
- Poisson regression, 71, 75, 218, 219, 245
 - estimate, 74, 114, 123, 127, 196, 233
- Poisson selection model, 170, 174
- police data, 93, 95, 112, 226, 227, 249
- positive regression dependence, 50, 69
- posterior distribution, 2, 3, 12, 229
- posterior interval, 12, 13, 232
- posterior probability, 18, 53, 57, 74, 95
- power, 34, 48, 57, 74, 83, 85–89, 104, 153, 163, 170–173
 - diagnostics, 83, 87, 88

- prediction, xi, 1, 7–10, 28, 211–217, 223, 228, 229, 234, 235, 237–241
 - computer-intensive, 212
- prediction error, 7, 214, 219–221
- prediction rule, 97, 211–213, 216, 217, 219–221, 223
 - ideal, 213
- prior density, 2, 27, 95–97, 228
- prior knowledge, 80, 88
- prior probability, 17, 70, 83, 187, 211, 213
- prostate data, 15, 17, 21, 24, 27, 41, 42, 44, 47, 50, 57, 60–63, 68, 71–73, 78–80, 82, 84, 85, 88, 148, 151, 184, 185, 207, 208, 212, 214–216, 219, 221, 224, 225, 229, 249, 250
- randomization, 164–167, 169, 174, 176, 179
 - row, 165, 167–171, 174, 177, 183, 184
- ras pathway, 164, 166, 171
- relevance, x, 81, 82, 94, 185, 186, 196–198, 205, 206, 209
- relevance function, 14, 197, 198, 209
- restandardization, 168, 169, 173, 174, 176, 178, 179, 183, 184
 - restandardized, 168, 173, 184
- risk function, 3
- rms approximation, 121–123, 125, 161
- Robbins, Herbert, x, 1, 14, 88, 240, 241
- SAM, 61–64, 94, 110, 174, 209
- scaling properties, 80, 81
- score function, 171, 246
- scoring function, 165, 169, 170, 174
- secant, 19
- second derivative matrix, 246
- selection bias, 215, 230
- separate analyses, 185, 186, 199, 205
- separation, 184–186, 196, 199, 200, 205, 209
- Shakespeare data, 238, 239, 241, 250
- Shakespeare, Wm., 238–241
 - canon, 238, 239
- Šidák procedure, 36, 37
- significance level, 34, 81, 178, 200
- Simes' inequality, 39
- simulation, 7, 8, 55, 56, 76, 87, 104, 106, 121, 146, 148, 149, 155, 173, 196, 217, 232
 - experiment, 9, 56, 107, 125, 126
- simz, 148, 159, 162, 250
- single nucleotide polymorphism (snp), 81, 147, 162, 202–206, 210
- singular value decomposition (svd), 118, 142
- skewness, 132, 135
- slope, 19, 34, 110, 194, 225, 233
 - statistic, 194
 - test, 152, 205
- small subclasses, 192
- smoothed cdf estimate, 72
- soft thresholding rule, 215, 218, 219
- spherical cap, 65–67
- spike-in, 207
- squared error loss, 3, 8
- standard deviation, 24, 55, 56, 76, 77, 87, 104, 109, 113–117, 122–126, 128–130, 132–135, 137–139, 143–146, 160, 165–168, 184, 195, 196, 217, 236
 - binomial, 114, 115, 122
- standard error, 16, 96, 130, 138, 156, 194, 202, 208, 219, 221
- standard normal, 16, 18, 32, 43, 53, 59, 60, 71, 72, 91, 116, 133, 136, 173, 244
- standardization, 91, 121, 144, 146, 148, 149, 151, 169
 - column, 144, 145, 152, 250
 - double, 142–147, 149, 151–153, 155, 162, 209, 250
 - quantile, 153
 - row, 144, 145, 169
 - row and column, 144, 145
 - standardized, 132, 135, 146, 169, 182, 183, 213, 221, 222, 249
- Stein, Charles, x, 1, 5, 7, 14, 28, 241
- step-down procedure, 37, 39
- step-up procedure, 39, 42, 49, 50, 81
- stepwise algorithms, 37
- strata, 200–202
- strong control, 35
- Student-*t*, 16, 33, 51, 65, 68, 105, 108, 112, 132
 - effects, 219, 221
- sufficient statistic, 243–245, 247
- t*-statistic, 64, 98, 110, 111, 133, 157, 206–209, 219, 221
 - two-sample, 16, 34, 41, 94, 105, 106, 108, 111, 125, 176, 214
- t*-test, x, 28, 32, 64–66, 69, 91, 96, 112, 149, 163, 168, 249

- tail counts, 160, 161
- tangent, 19, 23, 226, 227
- testing, x, xi, 1, 14–17, 34–36, 46, 57, 68, 88–90, 126, 134, 150, 152, 170, 172, 173, 194, 198, 208, 209, 240, 243
 - hypothesis, x, xi, 1, 2, 10, 14, 15, 27, 28, 30, 32, 58, 63, 70, 97, 110, 113, 141, 225
 - large-scale, 1, 15, 18, 25, 26, 34, 37, 42, 46, 48, 51, 70, 89, 90, 105, 109, 198, 225, 229
 - multiple, xi, 17, 30, 43, 45, 46, 48, 51, 57, 58, 70, 205
 - significance, 30, 37, 46, 48, 51, 70, 89, 227, 229
 - simultaneous, 15, 27, 28, 30, 64, 67, 164, 227
 - single-case, 47, 51, 55, 57, 70
- testing algorithm, 30, 34, 37, 45, 48, 51
- theoretical null, 17, 20, 21, 51, 60, 61, 75, 84, 89–97, 105–108, 110–112, 130, 141, 149, 169, 194, 202, 223, 225–227, 250
 - distribution, 16, 20, 32, 33, 81, 89, 90, 105, 106, 168, 185, 190, 223
- trace, 143
- training set, 211–213, 216
- transcription factor, 163
- trapping period, 234, 235, 237
- true and false discoveries, 46
- true discoveries, 47, 55, 59, 232
- Turing, Alan, 14, 241
- two-groups model, 17, 20, 21, 24, 27–29, 48, 53, 60, 70, 76, 78, 81–84, 98, 103, 187, 197, 211, 217, 227, 228
- Type I error rate, 30, 47
- voxels, 26, 34, 54, 57, 73, 181, 183, 185, 186, 192–194, 197
- weak control, 35
- Westfall and Young, 39, 40, 45
- Wilcoxon test, 28, 98
- wing-shaped function, 161
- Wishart distribution, 155, 156
- z-values, 16–18, 20–22, 26, 29, 31–34, 43, 44, 47, 51, 55, 56, 59, 60, 62–64, 69, 71, 88, 90, 91, 96, 99, 103, 104, 106, 110, 111, 114, 115, 125, 132, 134, 145, 149, 159, 160, 163–165, 167–169, 171, 173, 174, 177, 181, 184, 185, 190, 192–194, 196, 198, 200, 202, 205–207, 217–219, 245, 247, 249
 - correlated, 82, 148, 174, 250
 - histogram of, 34, 64, 71, 72, 78, 84, 90–94, 105, 106, 139, 164, 167, 182, 183, 193, 202, 204–206
 - non-null distribution of, 115, 132, 137
 - vector of, 61, 109, 113, 148, 159, 175, 219, 249, 250
 - zero assumption, 60, 87, 98–100, 102, 227